

Naoki Sato

PH.D. CANDIDATE IN COMPUTER SCIENCE

Meiji University, 1-1-1 Higashimita, Tama-ku, Kawasaki-shi, Kanagawa 214-8571 Japan

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Education

Meiji University

PH.D. CANDIDATE IN COMPUTER SCIENCE

- Topic: Optimization in Deep Neural Networks
- Supervisor: Prof. Hideaki Iiduka

Kanagawa, Japan

Apr. 2025 -

Meiji University

M.S. IN COMPUTER SCIENCE

- Thesis: Implicit Graduated Optimization with Noise in Stochastic Gradient Descent
- Supervisor: Prof. Hideaki Iiduka

Kanagawa, Japan

Apr. 2023 - Mar. 2025

Meiji University

B.S. IN COMPUTER SCIENCE

- Thesis: Using Constant Learning Rate of Two Time-Scale Update Rule for Training Generative Adversarial Networks
- Supervisor: Prof. Hideaki Iiduka

Kanagawa, Japan

Apr. 2019 - Mar. 2023

Research Interests

Optimization for Deep Learning

Publication

INTERNATIONAL CONFERENCE

Explicit and Implicit Graduated Optimization in Deep Neural Networks

[Naoki Sato](#), Hideaki Iiduka

The 39th Annual AAAI Conference on Artificial Intelligence (AAAI-25)

Philadelphia, Pennsylvania, USA

Feb. 27, 2025

Theoretical Analysis of Two Time-Scale Update Rule for Training GANs

[Naoki Sato](#), Hideaki Iiduka

The 10th International Congress on Industrial and Applied Mathematics (ICIAM2023)

Waseda University, Tokyo, Japan

Aug. 23, 2023

Existence and Estimation of Critical Batch Size for Training Generative Adversarial Networks with Two Time-Scale Update Rule

[Naoki Sato](#), Hideaki Iiduka

The 40th International Conference on Machine Learning (ICML2023)

Honolulu, Hawaii, USA

Jul. 24, 2023

JOURNAL

Scaled Conjugate Gradient Method for Nonconvex Optimization in Deep Neural Networks

[Naoki Sato](#), Hideaki Iiduka

Journal of Machine Learning Research (JMLR)

Dec. 13, 2024

DOMESTIC CONFERENCE (JAPAN)

Analysis of Muon's Convergence and Critical Batch Size

[Naoki Sato](#), Hiroki Naganuma, Hideaki Iiduka

The 2025 Fall National Conference of Operations Research Society of Japan

Hiroshima University, Hiroshima,

Japan

Sep. 12, 2025

Stochastic Frank Wolfe method for Constrained Nonconvex Optimization and its Application for Adversarial Attack

[Naoki Sato](#), Hideaki Iiduka

The 2025 Spring National Conference of Operations Research Society of Japan

Seikei University, Tokyo, Japan

Mar. 6, 2025

Global Optimization for Empirical Risk Function by Graduated Optimization with Stochastic Noise in SGD

Naoki Sato, Hideaki Iiduka

The 55th Information-Based Induction Sciences and Machine Learning (IBISML)

Hokkaido University, Hokkaido, Japan

Dec. 21, 2024

Global Optimization for Empirical Risk Minimization Problems using a Graduated Optimization Algorithm with the Smoothing Effect of Stochastic Gradient Descent

Naoki Sato, Hideaki Iiduka

The 27th Information-Based Induction Sciences Workshop (IBIS2024)

SONIC CITY HALL, Saitama, Japan

Nov. 5, 2024

Role of Momentum in Smoothing Objective Function and Generalizability of Deep Neural Networks

Naoki Sato, Hideaki Iiduka

The 2024 Fall National Conference of Operations Research Society of Japan

Nanzan University, Aichi, Japan

Sept. 11, 2024

Global Optimization of Deep Neural Networks for Graduated Optimization Method using Smoothing Effect of Stochastic Gradient Descent

Naoki Sato, Hideaki Iiduka **[Student Excellent Presentation Award]**

The 2024 Spring National Conference of Operations Research Society of Japan

University of Tsukuba, Ibaraki, Japan

Mar. 7, 2024

Existence and Estimation of Critical Batch Size for Training GANs with Two Time-Scale Update Rule

Naoki Sato, Hideaki Iiduka

RIMS Workshop on Mathematical Optimization: Theory and Practice, Research Institute for Mathematical Sciences

Kyoto University, Kyoto, Japan

Aug. 28, 2023